MATHEUS ANDRELL

Backend Software Engineer | Java, Spring Boot, Node.js, TypeScript | AWS | Event-Driven Architecture | Generative AI (RAG, MCP)
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SUMMARY

Backend Software Engineer focused on Java/Spring Boot, Node.js/TypeScript, event-driven architecture, messaging and AWS. Experience with microservices, asynchronous processing (SQS, Lambda, RabbitMQ), REST APIs, payment and ERP integrations, CI/CD and observability. Applies generative AI (RAG, MCP, Amazon Bedrock) to real business workflows. Building backend systems since 2023.

WORK EXPERIENCE

Software Engineer (Mid-level) | VMarket Food Solutions

Mar 2026 – Present | São Paulo, Brazil (Hybrid)

- Backend engineering for a B2B e-commerce and procurement platform for the food service industry, in a team of 5 developers (Node.js, TypeScript, NestJS, PHP/Laravel, AWS).
- Co-developed (in a pair) the MCP (Model Context Protocol) layer of a RAG-based quotation automation on Amazon Bedrock, cutting manual effort by more than 80% (month-over-month comparison, before vs. after rollout).
- Implemented automated matching between electronic invoices (NF-e) and purchase orders using Elasticsearch, in a workflow of approximately 6,000 invoices per day.
- Built an asynchronous pipeline with Amazon SQS, AWS Lambda and DynamoDB processing approximately 60,000 messages per day, with idempotency, dead-letter queues (DLQ) and peak-load handling.
- Reworked email delivery into an asynchronous workflow (SQS, Lambda, Serverless Framework) and implemented Redis caching with event-driven invalidation to cut redundant calls to AI services.
- Build REST APIs and microservices with third-party integrations; take part in code reviews and architecture discussions; troubleshoot production incidents with Grafana and structured logging (Pino, Monolog).

Junior Software Engineer | Evence Telecom

Oct 2024 – Feb 2026 | São Paulo, Brazil (Hybrid)

- Built REST APIs and enterprise web applications with Java, Spring Boot and Node.js in an event-driven microservices architecture.
- Implemented asynchronous service-to-service communication with RabbitMQ and integrated the Evolution API (WhatsApp) with the messaging infrastructure.
- Ran applications on AWS (EC2, S3, Lambda, SQS) with Docker and Kubernetes; implemented unit tests and observability with Grafana, Loki and Zabbix.
- Refactored the Vue.js component architecture using Atomic Design, reducing maintenance complexity by about 70%.

Junior Software Engineer | Dinamk Tech

Mar 2024 – Oct 2024 | São Paulo, Brazil (On-site) | Promoted from intern to junior after 3 months

- Built REST APIs and microservices with Java and Spring Boot for an academic research platform.
- Integrated the Iugu payment gateway, contributing to a 40% increase in completed payments, and implemented integrations with TOTVS ERP.
- Automated CI/CD pipelines with GitHub Actions for AWS deployments, replacing the manual FTP-based process (90% gain in deployment process efficiency).
- Took part in architecture decisions, technical solution design and code reviews.

Software Development Intern | Dinamk Tech

Jan 2024 – Mar 2024 | São Paulo, Brazil

- Contributed to migrating legacy applications from Java 8 to Java 17; built REST APIs with Spring Boot, JPA and Hibernate.
- Optimized SQL queries, refactored legacy code and implemented tests with JUnit and Mockito.

Freelance Software Engineer | Mentat Solutions

Jan 2024 – Jul 2024 | Remote | Part-time, in parallel

- Built a generative AI solution that analyzes and consolidates multiple documents, making the documentation process 60% more efficient.
- Built REST APIs with Java, Spring Boot and PostgreSQL, Python services for information extraction and React interfaces, with Docker and CI/CD.

IT Intern (Software Development & Support) | Grupo Educacional UNIS

Feb 2023 – Jan 2024 | Varginha, Brazil | Part-time

- Built REST APIs with Java and Spring Boot, optimized PostgreSQL queries and implemented tests with JUnit and Mockito.

- Provided first-level technical support; fixes for recurring failures contributed to about a 70% reduction in support tickets.

PROJECTS

EventCore | github.com/MatheusAndrell/eventcore
- Open source event ingestion and signed webhook delivery platform built with Java 25, Spring Boot, PostgreSQL and RabbitMQ.
- Transactional outbox with publisher confirms, idempotency (Idempotency-Key, RFC 8785), at-least-once delivery with retries, DLQ and replay.
- Security with HMAC-SHA256 (Standard Webhooks), AES-256-GCM and SSRF protection; observability with Prometheus and OpenTelemetry; tests with Testcontainers and Toxiproxy.

TECHNICAL SKILLS

Languages and frameworks: Java, Spring Boot, Spring Data, JPA/Hibernate, Node.js, TypeScript, NestJS, Express, PHP, Laravel, Python
Cloud and DevOps: AWS (Lambda, SQS, SNS, DynamoDB, EC2, S3, CloudWatch), Serverless Framework, Docker, Kubernetes, GitHub Actions, CI/CD
Architecture and messaging: Microservices, Event-Driven Architecture, RabbitMQ, Amazon SQS, REST APIs, Clean Architecture, DDD, SOLID
Data: PostgreSQL, MySQL, MongoDB, DynamoDB, Redis, Elasticsearch
Generative AI: RAG, MCP (Model Context Protocol), Amazon Bedrock, LLMs
Observability and testing: Grafana, Loki, Zabbix, Prometheus, OpenTelemetry, JUnit, Mockito, Testcontainers

EDUCATION

Postgraduate Specialization in Software Architecture | FIAP | Aug 2026 – Jun 2027 (in progress)
Associate Degree in Systems Analysis and Development (Tecnólogo) | Centro Universitário Eniac | Aug 2022 – Jan 2025

CERTIFICATIONS AND COURSES

Microservices in practice: Java and Spring; messaging with RabbitMQ; IaC with CDK and AWS deployment (Alura) | Java Architecture: Clean Architecture and scalable infrastructure (Alura) | AWS Observability with CloudWatch (Alura) | Manage Kubernetes in Google Cloud Skill Badge (Google) | Managing Cloud Infrastructure with Terraform (Google) | Fundamentals of Backend Engineering (Udemy)

LANGUAGES

Portuguese: native | English: advanced